

Lean 4 Stokes Blueprint

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Chapter 1

Stokes formalization blueprint

This blueprint records the current public proof architecture of the Lean 4 Stokes project. It is written directly from the latest first-principles entry point.

The current public theorem is the first-principles represented Stokes theorem in the `CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput` namespace. It proves a compact-support represented Stokes equality. The public input is only chartwise smoothness of the form and compactness of the closed algebraic support. The finite chart-box cover, pointwise half-space boxes, support-controlled partition, refined boundary boxes, zero-localized support proofs, local half-space Stokes fields, and represented endpoints are generated internally.

Green nodes in the figures are implemented Lean declarations. Blue nodes are the remaining bridge from represented integrals to a future mathlib-native manifold integral theorem.



Figure 1.1: Current first-principles compact-support represented Stokes route.

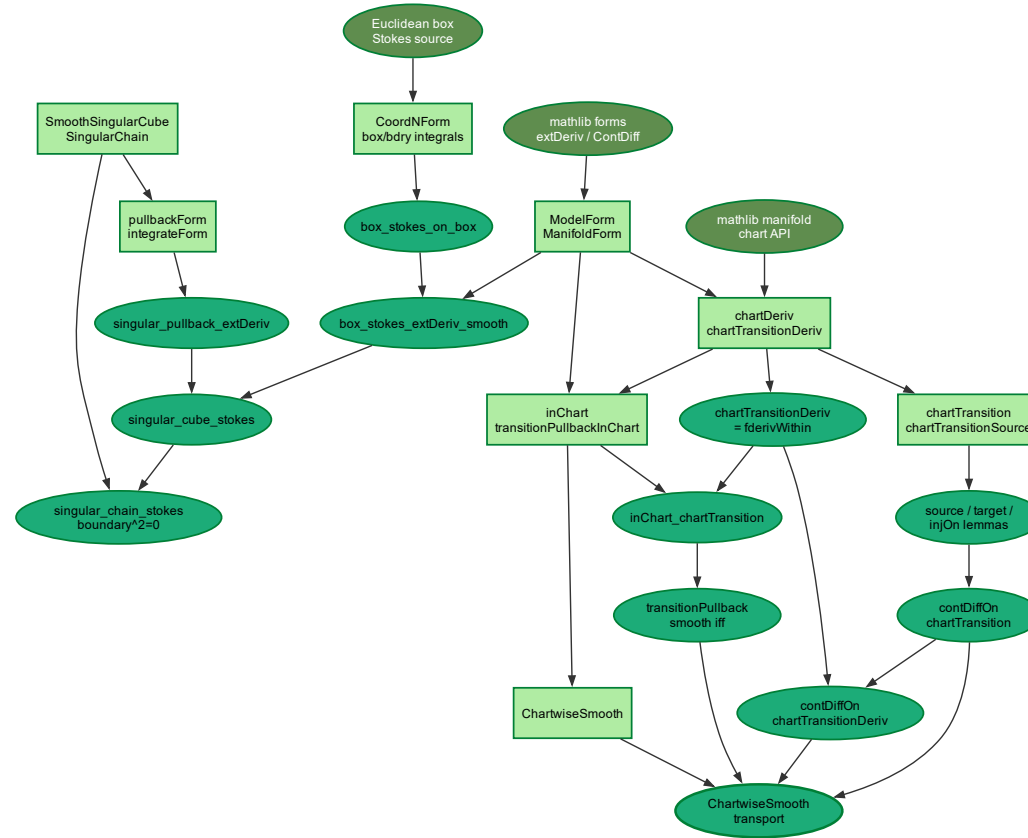


Figure 1.2: Core Euclidean, singular-cube, and chart-representative API.

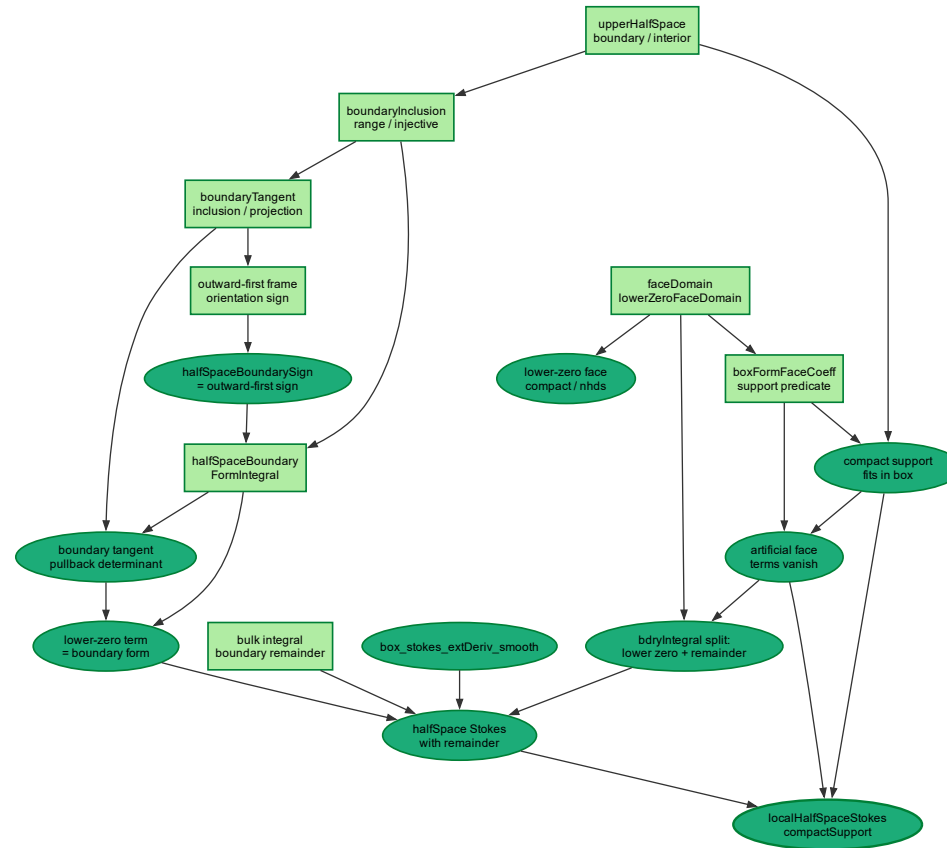


Figure 1.3: Half-space local Stokes and artificial-face cancellation.

1.1 Current Public Theorem

Definition 1.1 (First-principles input).

The preferred public input for the represented compact-support theorem. For a form ω , it exposes only:

`chartwiseSmooth : ManifoldForm.ChartwiseSmooth I omega`

and

`compactSupport : IsCompact (closure (ManifoldForm.support I omega)).`

The compact carrier is definitionally the closure of the algebraic support.

Definition 1.2 (Canonical represented endpoints).

The bulk and boundary sides are generated from the canonical intrinsic route. They are represented integrals: finite sums of local coordinate integrals over selected half-space boxes and their true boundary faces.

Theorem 1.3 (First-principles represented Stokes).

The canonical represented bulk integral equals the canonical represented boundary integral:

`canonicalRepresentedBulkIntegral = canonicalRepresentedBoundaryIntegral.`

This is the current main theorem exposed by `Stokes.Global.CoverIndexedFirstPrinciples`.

Definition 1.4 (Thin public import).

The public cover-indexed API imports only the first-principles theorem route. The default public surface is therefore the compact-support represented Stokes theorem rather than intermediate certificate constructors.

1.2 Generated Geometry

Definition 1.5 (Half-space model).

The model-with-corners range is identified with the upper half-space. This gives Lean the coordinate fact needed to decide whether a self chart point is an interior point or a boundary point.

Theorem 1.6 (Self-chart pointwise boxes).

For every point of the compact support carrier, self charts generate either an interior chart box or a boundary half-space box. At boundary points the lower normal corner is zero; at interior points the box lies in the positive half-space.

Definition 1.7 (Intrinsic input generated from first principles).

The first-principles input constructs the intrinsic input whose fields are pointwise chart-box data, compactness of the carrier, chartwise smoothness, and support containment in the carrier.

Definition 1.8 (Open selected cover and partition).

Compactness of the support carrier and the pointwise chart-box neighborhoods generate a finite selected cover and a support-controlled partition.

Definition 1.9 (Selected boundary finite half-space cover).

Boundary active coordinate carriers are covered by finitely many selected half-space boxes. These boxes are the local domains used by the boundary half-space Stokes theorem.

Definition 1.10 (Selected smooth refinement).

The selected finite cover is refined by smooth coefficients. The refined boundary pieces are definitionally aligned with the selected boundary finite cover.

1.3 Zero-Localized Local Stokes

Definition 1.11 (Zero-localized representative).

Chart representatives are totalized by zero outside the relevant source and target domains. This avoids false support claims for ordinary total chart representatives.

Theorem 1.12 (Support of selected zero-localized pieces).

The zero-localized transition pullback of each selected refined localized form has topological support contained in the corresponding half-space support box. This is the support fact that kills artificial faces in the local box theorem.

Definition 1.13 (Interior Stokes fields from chartwise smoothness).

Chartwise smoothness and the smooth selected refinement construct the analytic fields required by the half-space box theorem for each selected refined local piece.

Theorem 1.14 (Intrinsic route constructor).

The selected smooth refinement, the zero-localized support theorem, and the generated interior fields produce the complete intrinsic route certificate.

Theorem 1.15 (Intrinsic represented Stokes).

The intrinsic input proves the represented Stokes equality before the first-principles wrapper hides the pointwise chart-box carrier.

1.4 Euclidean And Half-Space Core

Definition 1.16 (Model and manifold forms).

A model form is a coordinate differential form, and a manifold form is a fiberwise alternating map on tangent spaces.

Theorem 1.17 (Box Stokes).

The coordinate Euclidean box theorem is repackaged for mathlib-style forms and `extDeriv`.

Definition 1.18 (Half-space support box).

Half-space support boxes encode the local geometry that separates the true boundary face from artificial box faces.

Theorem 1.19 (Half-space local Stokes).

Local Stokes on a half-space box reduces the boundary of the box to the true lower normal face; artificial faces vanish by support control.

Definition 1.20 (Boundary sign convention).

The half-space boundary sign is aligned with the outward-normal-first convention used by boundary chart integrals.

1.5 Remaining Native-Integral Bridge

Theorem 1.21 (Represented-to-native bulk integral).

Proposed Lean name: *Stokes.compareRepresentedBulkIntegralToMathlib*.

This future theorem should identify the canonical represented bulk integral with a mathlib-native integral of $d\omega$, once the target manifold-form integral API is available or chosen.

Theorem 1.22 (Represented-to-native boundary integral).

Proposed Lean name: *Stokes.compareRepresentedBoundaryIntegralToMathlib*.

This future theorem should identify the canonical represented boundary integral with the induced outward-first boundary integral of ω .

Theorem 1.23 (Mathlib-native compact-support Stokes).

Proposed Lean name: *Stokes.globalStokes_compactSupport*.

After both comparison theorems are available, the represented theorem becomes the native compact-support Stokes statement

$$\int_M d\omega = \int_{\partial M} \omega.$$

Theorem 1.24 (Compact manifold Stokes).

Proposed Lean name: *Stokes.globalStokes_compactManifold*.

The compact-support theorem specializes to compact oriented smooth manifolds with boundary by taking the support carrier to be the whole compact manifold.

1.6 Artifact Checks

The current project verification gates are:

```
lake build Stokes.Global.CoverIndexed
```

and

```
lake build Stokes.
```

The no-placeholder scan is:

```
rg "\bsorry\b|\badmit\b|^s*axiom\b" --glob "*.lean"
```

The theorem-level audit target is:

```
#print axioms
Stokes.CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.representedStokes
```

The expected result is the standard Lean/mathlib axiom set:

```
[propext, Classical.choice, Quot.sound].
```